

**Design, Development and Evaluation of a Cryptographically Secure
Document Tracking System with Structured Adaptive Hash-Chains and
Ed25519 Non-Repudiation for DepEd Iligan City**

A Thesis

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CHAPTER 1

THE PROBLEM AND ITS BACKGROUND

1.1 Introduction

In the modern landscape of public administration, Document Tracking Systems (DTS) serve as the essential digital infrastructure for monitoring the lifecycle of official correspondence. A DTS is a specialized information system designed to record the intake, movement, and final disposition of physical and digital files within an organization. For large-scale entities like the Department of Education (DepEd) Division of Iligan City, these systems are no longer optional tools; they are foundational to ensuring that personnel records, financial requests, academic reports, and other official correspondence are processed in an orderly and transparent manner. By providing a centralized view of where a document is and who is currently handling it, these systems transform abstract administrative processes into visible, manageable workflows (Frankel et al., 2018).

Before the introduction of digital tracking, government offices relied almost exclusively on manual logbooks and physical routing slips. In this traditional setup, a document's journey was recorded by hand at every desk it reached, often requiring staff to flip through thick paper ledgers to find a specific entry which over time would increase to thousands. While this method was sufficient for lower volumes of work, it was highly susceptible to human error, physical damage, and loss (Ozuruonyeoke et al., 2025). There was no way to search for a document instantly, and the lack of a centralized backup meant that if a logbook was lost, the entire history of those documents disappeared with it. The transition to digital DTS models was driven by the need for speed, searchability, and the ability to manage thousands of records without the physical limitations of paper. On a global scale, this modernization aligns directly with the United Nations' Sustainable Development Goal 9 (SDG 9)—Industry, Innovation, and Infrastructure—by establishing resilient, equitable, and sustainable digital public infrastructure that fosters institutional innovation (United Nations, 2015).

The primary goal of a DTS is to eliminate the uncertainty and lack of accountability often found in manual workflows. Without a tracking system, documents frequently enter a state where their status and location are unknown to both the submitter and the management until they reach their destination. This leads to systemic delays, misplaced files, and a phenomenon known as Ghost Documents, which are requests that move through the office without any official record (Ozuruonyeoke et al., 2025). Most DTSs aim to solve these

issues by providing a real-time audit trail, ensuring that every hand-off is timestamped and every delay is identifiable. This digital accountability is particularly critical under laws like Republic Act No. 11032 of the Philippine Government, which mandates that government actions be time-bound and verifiable to prevent bureaucratic slow-downs (Republic Act No. 11032, 2018).

This thesis project differs from most standard DTSs by moving beyond simple database logging into the realm of *Mathematical-Based Security* or *Cryptography*. While traditional systems rely on the hope that administrators will not change data (Policy-Based Security), this project implements a *Trust Builder* module using SHA-256 hash-chaining and Ed25519 digital signatures. Every log entry is mathematically linked to the one before it, making it impossible to secretly alter the history of a document. Furthermore, by requiring a Security PIN to generate cryptographic signatures for critical actions such as confirming that a document's processing is finished or adding a document to the dashboard, the system ensures non-repudiation—a user cannot later deny that they performed a specific action. A comparative analysis of related systems in Chapter 2 confirms that most existing local and foreign document tracking systems rely exclusively on policy-based access controls, leaving them structurally vulnerable to unauthorized, subtle alterations to data and the bypassing of security controls that this project is designed to prevent (Bajoria, 2025; Crosby & Wallach, 2009).

For IT students, this study is a vital bridge between theoretical classroom concepts and the practical demands of the software industry. Developing this system allows the researchers to apply advanced topics—such as cryptographic salt derivation, database indexing for millions of records, and asynchronous background processing—to solve a tangible community problem. Beyond technical skills, the project teaches the importance of Security by Design and the ethical responsibility of a developer to protect the integrity of government data. By building a system that directly impacts the efficiency of the DepEd Division Office, the researchers deepen their understanding of how information technology serves as a practical instrument for governance, transparency, and institutional accountability.

1.2 Background of the Study

The origin of this study began with an empirical observation visit to the DepEd Division Office of Iligan City in December 2025, facilitated by the official responsible for the researchers' survey authorization for a prior study. During this visit, researchers witnessed a *race condition* at the guest intake kiosks. Upon reporting these observations to the IT and

Records Departments, they confirmed that the current DTS lacked robust queue management, causing concurrent submission attempts to result in a race condition (Silberschatz et al., 2020). This failure meant that one guest's print job would often overwrite another's if they happened to press submit at almost the same time, leaving a guest with incorrect tracking information and the need to submit the same request again.

Beyond the visible failure at the kiosks, deeper consultation with the Records and IT Departments revealed that the same DTS was not being utilized to its full potential. Specifically, a significant proportion of internal documents — empirically observed to represent the majority of daily intake — frequently bypassed the initial digital submission process, resulting in a systemic lack of Tracking Codes—the primary keys essential for the system's database integrity. To manage these unrecorded documents and ensure operational continuity, the Records Department established a *Shadow System* (Behrens, 2009; Šedivcová & Potančok, 2019) using Google Forms and Google Sheets. Staff would manually transcribe document metadata—such as client names and purposes—from printed forms into Google Sheets to maintain a searchable record that the main DTS could not provide for all documents.

This fragmentation extends into the releasing phase of documents, where another shadow system is employed. Because the legacy DTS permits documents to bypass the initial digital intake, internal documents often lack a system-generated tracking code. To manage this lack of end-to-end traceability, the releasing department generates independent *releasing codes* for formality and documentation; empirical observation revealed a database exceeding 10,000 manually inputted entries. This practice, while providing a temporary solution for the staff, results in a disconnected and technically flawed database structure where the lifecycle of some documents cannot be reliably traced from intake to release. On observation, the reliance on unencrypted Google Sheets for over 10,000 entries represents a significant risk to data integrity, as these files lack the ACID compliance and mathematical security needed to prevent unauthorized or accidental modifications (Haag & Eckhardt, 2024).

When comparing this situation to other standard DTSs used in the public sector, a clear gap in functionality emerges. Most existing DTS solutions are designed to handle either only external guest intakes or only internal staff workflows, but rarely both under a unified tracking architecture (Acedo et al., 2025). In the case of DepEd Iligan, previous digital attempts failed because they did not account for the high volume of internal correspondence that bypasses the formal intake process. While other systems may have digitized the hand over process, they failed to provide the mathematical assurance that those logs remained

untampered over time. The institution was essentially relying on Policy-Based Security—the hope that staff would follow the rules—rather than Mathematical-Based Security that makes rules impossible to break (Bajoria, 2025).

To bridge the 'Physical-to-Digital' gap, the system enforces a digital gatekeeping mechanism. Subsequent departments are programmatically restricted from 'Receiving' a document unless it possesses a valid cryptographic signature from the previous station. This protocol ensures that physical movement without a QR scan results in a state of digital blockage — where further departmental processing is programmatically halted until a valid cryptographic scan is recorded, effectively mandating the use of the software for all administrative progress.

Following these revelations and the recommendation of the IT Head, the researchers recognized that a standard, off-the-shelf tracking system would not solve the problem. The study therefore pivoted to create a custom, high-security prototype tailored to the unique workflow of DepEd Iligan City. This solution is specifically designed to meet the IT Department's strict constraints: it must be built using PHP and restricted to a local area network (LAN) to ensure data sovereignty and eliminate exposure to external threat actors, who are responsible for 65% of all data breaches according to the 2024 Verizon Data Breach Investigations Report (Verizon, 2024). To directly resolve the kiosk race conditions, the system implements strict database-level concurrency controls and queue management. Furthermore, to address the distinct problem of data integrity, the system integrates hash-chaining and Ed25519 signatures. This ensures the proposed system is not just another tracking tool; it is a specialized security artifact designed to restore institutional trust and provide a permanent solution to the proliferation of informal, insecure tracking methods.

1.3 Statement of the Problem

The primary challenge facing the DepEd Division Office of Iligan City is the technical instability and incomplete operational integration of its existing DTS, which results in data corruption and the proliferation of insecure administrative practices. A critical manifestation of this instability is the occurrence of race conditions within the guest intake kiosks, where concurrent submissions often result in simultaneous print jobs overwriting one another (Silberschatz et al., 2020). This technical flaw leads to the immediate loss of tracking data for citizens, creating a race condition that undermines institutional reliability and prevents the system from maintaining a consistent record of external requests.

Beyond these visible failures, the organization suffers from a systemic fragmentation of its tracking architecture due to the operational bypassing of internal correspondence. A volume of documents frequently skips the formal digital intake process, resulting in a database that lacks the mandatory tracking codes—the primary keys—essential for end-to-end traceability. To compensate for these gaps and ensure operational continuity, the Records Department has resorted to the use of *Shadow Systems* (Behrens, 2009; Šedivcová & Potančok, 2019) using unencrypted platforms like Google Forms and Google Sheets. While these informal methods provide a temporary solution for staff, they introduce severe data integrity risks and ACID compliance necessary to protect sensitive government records (Haag & Eckhardt, 2024).

This lack of a unified tracking protocol further destabilizes the final stages of the document lifecycle. Because the legacy DTS permits documents to bypass the initial intake record, the releasing department is forced to generate independent releasing codes for the sake of formality. Direct observation reveals this practice creates a disconnected and technically flawed database structure where some documents cannot be reliably traced from intake to final release. Consequently, the institution is unable to provide a verifiable audit trail, leading to the proliferation of Ghost Documents Ozuruonyeoke et al. (2025) and failing to meet the accountability and transparency mandates of the Ease of Doing Business and Efficient Government Service Delivery Act (Republic Act No. 11032, 2018). Additionally, the absence of any intelligent routing assistance forces Records Officers to manually determine the appropriate departmental path for each document, a repetitive cognitive task that contributes to processing delays and human error in high-volume workflows.

1.4 Objectives of the Study

General Objective:

To design, develop, and evaluate a cryptographically secure DTS for DepEd Iligan City that ensures unified accountability and operational efficiency.

Specific Objectives:

1. Develop a Trust Builder module responsible for Audit Trail: Implement a SHA-256 hash-chaining mechanism utilizing a *Micro-sharding* architecture to ensure the immutability and scalability of document logs.

2. Integrate Ed25519 Non-Repudiation: Require digital signatures for critical administrative actions using elliptic curve cryptography to achieve Mathematical-Based Security.
3. Establish Unified Tracking and Heuristic Routing: Consolidate all document requests (internal and external) through a centralized guest portal to reduce code fragmentation, shadow system usage, and integrate a heuristic route prediction module using TF-IDF keyword weighting as a decision-support tool for Records Officers to automate repetitive routing decisions.
4. Optimize Physical Workflows: Integrate QR code scanning at every department hand-off to confirm as a physical receipt and delivery. This ensures the digital state matches the physical location.
5. Evaluate System Integrity and Usability: Assess the prototype's ability to detect tampering through an automated *Active Guard* module and measure usability using the System Usability Scale (SUS).

1.5 Significance of the Study

The findings and technical deliverables of this study will provide substantial operational and security benefits to the DepEd Division of Iligan City and the broader academic community. Specifically, the following groups will directly benefit from the implementation of this system:

- DepEd Records Department: The system mitigates the tedious and error-prone reliance on manual shadow systems (Google Forms and Sheets). By enforcing a mandatory intake protocol and providing an automated tracking architecture, it significantly reduces manual transcription and administrative fatigue, allowing staff to process documents with greater speed and accuracy.
- DepEd IT Department: The prototype provides the IT staff with a highly secure, PHP-based local solution that strictly adheres to the division's on-premise network (LAN) constraints. By replacing unencrypted shadow systems with a centralized database secured by SHA-256 hash-chaining, it drastically reduces the organization's vulnerability to insider threats and data corruption.
- DepEd Management and Division Officials: The integration of Ed25519 digital signatures and QR code scanning provides management with mathematical proof of action (non-repudiation) and verifiable timestamps. This ensures institutional

transparency and strict compliance with the accountability mandates of the Ease of Doing Business Act (Republic Act No. 11032).

- Guests and the General Public: Citizens submitting requests will experience a more reliable and professional digital service. The elimination of kiosk race conditions guarantees that their document data will not be overwritten, preventing frustration and ensuring real-time visibility into the status of their requests.
- Future IT Researchers: This study serves as a practical, documented reference for implementing advanced cryptographic features—specifically blockchain-inspired hash-chaining and Ed25519 signatures—within resource-constrained, centralized relational databases (RDBMS) in the public sector, and also to provide a template for DTSs.
- Global Sustainable Development (SDG 9): By replacing fragile manual processes with a high-security, cryptographically backed digital infrastructure, the system actively contributes to the targets of SDG 9. It positions the public sector not merely as a consumer of technology, but as a driver of resilient Digital Public Infrastructure (DPI) that guarantees data sovereignty and fosters technological innovation in local governance (Liashenko et al., 2024).

1.6 Scope and Delimitation

The study is geographically and operationally bounded within the DepEd Division Office of Iligan City. It covers the end-to-end document request lifecycle, beginning from the initial guest intake at the kiosks, through internal department-to-department transfers, and concluding with the final physical release of the document. The system serves as a digital synchronization layer and an immutable logbook, utilizing the DocumentLog architecture to record and verify the physical location and state of documents in real-time. The heuristic route prediction module suggests routes but does not autonomously force document transfers.

The project is strictly limited to a local area network (LAN) deployment as mandated by the institutional IT department to ensure data sovereignty. Technical implementation is restricted to the PHP programming language; specifically, the prototype is built using the Laravel 12 framework to provide a modern, optimized structure while adhering to the client's language constraints. Additionally, the system utilizes MySQL as the relational database engine, selected for its native support for row-level locking — a critical requirement for resolving concurrent transaction conflicts — its proven compatibility with the Laravel Eloquent ORM, and the researchers' demonstrable proficiency in MySQL database

architecture and optimization, which ensures that the implementation is both technically sound and efficiently executed (Silberschatz et al., 2020). Furthermore, the system is designed to digitize the tracking metadata and the security audit trail rather than replacing the physical storage or movement of original paper documents. It does not extend to the DepEd Regional Office or other external divisions.

1.7 Definition of Terms

For the purpose of this study, the following terms are defined to ensure a clear understanding of the technical concepts presented:

Term	Definition
ACID Compliance	A set of database properties (Atomicity, Consistency, Isolation, Durability) that guarantee that transactions are processed reliably, ensuring data integrity even in the event of system failures.
Active Guard Module	A specialized security middleware that verifies a document's current database state against its last signed record before authorizing any new action.
Auto-Freeze	An automated defensive mechanism that locks a document's record from further processing if the Active Guard module detects unauthorized data manipulation.
Digital Public Infrastructure (DPII)	Refers to shared, interoperable digital systems—such as digital identification, payment platforms, and data exchange mechanisms—that support the secure delivery of public and private services.
Ed25519	A high-performance, elliptic curve digital signature scheme used to provide cryptographic proof of identity for administrative actions.
Hash-Chaining	A cryptographic method where the hash of a previous log entry is included in the current entry, creating a tamper-evident audit trail where the history cannot be altered without breaking the chain.

Local Area Network (LAN)	A computer network that interconnects devices within a limited geographic area, such as a single office or building.
Micro-sharding	An architectural pattern that isolates the security hash-chains into document-specific segments, ensuring the system remains scalable and fast as the record volume grows.
Mathematical-Based Security	The security paradigm of the project, which relies on cryptographic proofs (like hash-chains and digital signatures) rather than human or administrative oversight (Policy-Based Security).
Non-Repudiation	A security assurance that provides proof of the origin and integrity of data, making it impossible for a user to deny having performed a specific action or signature.
Personally Identifiable Information (PII)	Any data that can directly or indirectly identify a specific individual, such as names, Social Security numbers, biometric records, or IP addresses.
Policy-Based Security	The contrasting security model where system integrity relies on administrative rules, access controls, and the assumption that personnel will not abuse their access to data.
Race Condition	A technical failure occurring when multiple processes attempt to access or modify shared data simultaneously, resulting in unintended behavior or data corruption.
Rapid Application Development (RAD)	An iterative software development methodology introduced by Martin (1991) that prioritizes continuous user feedback and incremental prototype refinement over rigid sequential planning.

SHA-256	A secure cryptographic hash function that generates a fixed-size, unique digital fingerprint of data, utilized in this study for constructing the hash-chain.
Shadow System	Unofficial IT solutions, such as manual Google Sheets or Forms, deployed by staff to bypass or supplement failing official systems, often leading to data fragmentation and security risks.
TF-IDF	Term Frequency-Inverse Document Frequency; a numerical statistic used in the route prediction module to weigh the importance of keywords in document purposes to assist in heuristic routing suggestions.
Tracking Code	A unique, system-generated alphanumeric identifier that serves as the primary key for a document, enabling its end-to-end traceability across the organization.
Trust Builder Module	The centralized software module responsible for calculating hashes, managing chains, and maintaining the mathematical integrity of the DTS.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a structured review of related literature and existing systems. It focuses on document tracking inefficiencies, cryptographic security, and administrative transparency. It concludes with the theoretical and conceptual frameworks that anchor the development of the DTS.

2.1 Related Literature

To establish a rigorous foundation for this study, the related literature is organized into six thematic pillars: administrative inefficiencies, cryptographic integrity, physical-digital synchronization, shadow system workflows, system scalability, and user-centric design frameworks.

2.1.1 Administrative Bottlenecks and the Failure of Manual Systems

The transition from physical to digital administration is driven by the persistent failure of manual methods to handle modern correspondence volumes. Manual document tracking represents the traditional *pen-and-paper* era, where the movement of files is recorded in physical logbooks and routing slips. While this method was historically sufficient for low-traffic environments, it is increasingly viewed as an obstacle to modern governance due to its lack of searchability, physical vulnerability, and systemic opacity (Acoba , 2024).

A synthesis of global and local literature reveals a consensus on the severe consequences of these manual bottlenecks. Researchers like Ozuruonyeoke et al. (2025) have quantified these failures, reporting that manual systems in local government units suffer from a 40% document misplacement rate and prolonged delays that average three to six weeks. Furthermore, Acoba (2024) and Maina et al. (2025) highlight that even in the Philippine judicial and administrative sectors, the reliance on official record books leads to *information silos* where accountability is impossible to enforce. While some studies focus on the technical loss of data (Nihare et al., 2020; Kolte et al., 2021), others emphasize the legal contradictions between manual slow-downs and the strict turnaround times mandated by Republic Act No. 11032 (Acedo et al., 2025).

These literatures help this research by providing the empirical justification for a centralized digital tracking solution. By documenting the specific failure points of physical logbooks—such as lost tracking codes and the *ghost document* phenomenon—these studies

allow the researchers to design a system that specifically targets institutional bottlenecks. The literature establishes that a successful DTS must go beyond simple digitization; it must implement mandatory intake protocols to prevent the operational bypassing that occurs in the manual workflows described by Agarkar et al. (2012) and Akçay et al. (2019).

2.1.2 Cryptographic Integrity and Tamper-Evident Architectures

The evolution of document security has shifted from traditional policy-based access levels toward mathematical-based assurance. In standard information systems, security is often policy-based, meaning it relies on the hope that administrators will not abuse their access to change historical records. However, modern cryptographic literature argues that for government data, security must be built directly into the data structure itself to ensure that the history of a document remains immutable and verifiable (Bajoria, 2025; Crosby & Wallach, 2009).

Synthesis of current research shows that tamper-evident logging and digital signatures are the foundational technologies for this shift. Chavarro et al. (2018) and Crosby and Wallach (2009) highlight the importance of foundational cryptographic concepts like history trees and hash-chaining, which remains the industry standard for tamper-evident logging. This architecture is further strengthened by the seminal work of Bernstein et al. (2012) on Ed25519 signatures, which provides high-speed, non-repudiable proof of action. While recent foreign studies focus on the decentralization of these chains via blockchain (Mishra & Ganesan, 2025; Zhai et al., 2022), there is a growing debate on how to implement such high-integrity ledgers within centralized databases optimized for local government hardware (Bajoria, 2025; Kim & Kim, 2024).

These literatures are vital to the development of the DTS Trust Builder module. They provide the mathematical frameworks—specifically SHA-256 hash-chaining and Ed25519 elliptic curve signatures—needed to create an immutable audit trail. By building upon these concepts, the researchers can ensure that the prototype provides a level of security that exceeds the capabilities of traditional government systems. The literature confirms that by shifting trust from human supervisors to mathematical proofs (Zhai et al., 2022), the system can effectively eliminate *silent tampering* and ensure institutional accountability.

2.1.3 Physical-Digital Synchronization via QR and IoT Technologies

Effective document tracking requires a seamless bridge between the physical movement of files and their digital records. Synchronization technology ensures that the

status displayed on a computer screen accurately reflects the ground truth of a document's physical location. Without this link, a tracking system becomes a mere repository of disconnected data, failing to provide the real-time visibility needed for administrative oversight (Acedo et al., 2025; Farin, 2024).

The synthesis of literature on synchronization identifies QR codes as the most practical and acceptable technology for public sector environments. Farin (2024) and Acedo et al. (2025) found that QR scanning is highly accepted in Philippine institutions due to its cost-effectiveness and ease of use. While more advanced technologies like NFC and IoT (Abidin et al., 2019, 2020) offer superior real-time localization, they are often seen as too expensive or complex for local government units. Conversely, studies on barcodes (Rellon et al., 2020) and watermarking (Rimando & Corpuz, 2022) suggest that while functional, they lack the rapid scanning and identification capabilities of modern QR tags (Tan et al., 2023).

These literatures helps the study by justifying the selection of QR code technology as the primary mechanism for department-to-department hand-offs. It establishes that for a system to be *active*, it must require a physical scan to move a document from an *in transit* to a *processing* state. This synchronization directly supports the requirements of RA 11032 by providing verifiable, timestamped proof of every transfer. Furthermore, by learning from the acceptability studies of Jaquilmo and Sarmiento (2023) and Hermosa et al. (2023), the researchers can design a scanning interface that minimizes administrative friction while maximizing physical accountability.

2.1.4 Shadow Systems and the Evolution of Unofficial Workflows

The emergence of Shadow Systems—unofficial software solutions deployed by staff to bypass failing official systems—represents a significant cybersecurity and data integrity risk. Shadow systems usually emerge as a rational coping mechanism when formal IT solutions are perceived as too rigid, slow, or disconnected from actual workflows (Behrens, 2009; Šedivcová & Potančok, 2019). In many government offices, this manifests as a reliance on unencrypted Google Sheets and Forms to manage critical document metadata (Muhammad, 2019).

A synthesis of the literature identifies a critical tension between operational speed and official security. Behrens (2009) and Rentrop and Zimmermann (2020) argue that Shadow Systems provide a human-centric flexibility that official systems often lack. However, Haag and Eckhardt (2024) and Zhai et al. (2022) warn that these unapproved systems lead to fragmented databases and severe vulnerabilities, as they lack the ACID compliance and

mathematical security needed for government records. This trend is worsening in the AI era, where staff can easily deploy consumer-grade tools that are faster than institutional software but significantly less secure (Haag & Eckhardt, 2024).

This literature helps the research by framing the DTS prototype not just as a security tool, but as a bespoke alternative to shadow systems. By recognizing that staff use Google Sheets for its speed and convenience (Haag & Eckhardt, 2024; Lingaya, 2019), the researchers can focus on developing features—such as smart routing and sub-second load times—that compete with the perceived benefits of informal tools. The literature informs the design philosophy of the project: to mitigate shadow systems, the official system must be more usable, more reliable, and faster than the informal alternatives it seeks to replace.

2.1.5 Performance Optimization and Scalability in High-Volume Systems

Systems designed for large government divisions like DepEd must be capable of handling millions of records without a degradation in performance. For a DTS, scalability is not just about storage; it is about the system's ability to perform complex cryptographic audits and search queries in sub-second response times (Silberschatz et al., 2020). High-volume environments create significant stress on centralized databases, requiring advanced architectural patterns to maintain institutional efficiency.

The synthesis of literature on system performance highlights a shift toward asynchronous processing and isolated data structures. Modern architectural patterns emphasize that traditional server-based models must incorporate background jobs to handle intensive tasks like report generation (Silberschatz et al., 2020). More importantly, Kim and Kim (2024) introduced a structured hash chain approach that isolates logs into document-specific chains. This micro-sharding technique ensures that the mathematical overhead of hash calculation does not slow down the entire system as the record volume grows. This technical consensus is supported by various engineering studies on the use of iterative models and modern frameworks (Aqeel et al., 2023; Lee & Paik, 2019; Rosidin et al., 2023).

These literatures help the study by providing the technical blueprint for the DTS prototype's performance layer. By incorporating asynchronous background processing (Thohir et al., 2022) and efficient search indexing (Krohn, 2019), the researchers can ensure the system remains responsive under heavy load. The work of Kim and Kim (2024) specifically informs the micro-sharding of the hash chain, allowing the system to verify individual document integrity without needing to scan the entire global ledger. This ensures

that the system meets the high-volume performance metrics defined in Chapter 3 while maintaining strict mathematical security.

2.1.6 Human-Centric Design and Technology Acceptance

The ultimate success of a technological artifact is determined by its acceptance and adoption by its human users. Even the most secure system will fail if it is perceived as too complex, culturally insensitive, or disconnected from organizational habits. Therefore, the design of a DTS must be grounded in theories that explain how technology integrates with human behavior and decision-making processes (Bertalanffy, 1968; Frankel et al., 2018).

A synthesis of human-centric literature reveals that technology acceptance is driven by perceived usefulness and the reduction of systemic uncertainty. Frankel et al. (2018) and Bertalanffy (1968) provide the theoretical foundations for viewing the office as a socio-technical system where technological tools must support, rather than disrupt, established habits. Research on institutional technology adoption consistently demonstrates that systemic stressors — such as information gaps, rigid workflows, or complex protocols — lead to resistance and workaround behaviors that undermine official systems (Behrens, 2009; Šedivcová & Potančok, 2019). Conversely, providing clear, data-driven features through Decision Support Systems (Keen & Scott Morton, 1978; Ngan et al., 2022) has been shown to improve engagement and trust (Christinna & Saragih, 2022; Khairi et al., 2022).

These literatures help the research by guiding the UI/UX design and the implementation of SMART (Specific, Measurable, Achievable, Relevant, and Time-bound) features. By utilizing Decision Support Theory, the researchers integrated an heuristic route prediction module to help Records Officers manage document routing efficiently. The literature on technology acceptance ensures that the cryptographic actions—such as PIN-based signing—are presented in a familiar, non-intimidating way. By prioritizing the “human factor” and institutional trust, the study aims to create a system that is not only mathematically secure but also culturally and operationally acceptable to the personnel of DepEd Iligan City.

2.1.7 Digital Public Infrastructure and Sustainable Development (SDG 9)

The modernization of e-governance systems is increasingly recognized not merely as an administrative upgrade, but as the construction of resilient Digital Public Infrastructure (DPI). This aligns with the United Nations' Sustainable Development Goal 9 (SDG 9), which emphasizes building robust infrastructure and fostering inclusive innovation (United Nations,

2015). Recent studies, such as those by Liashenko et al. (2024), demonstrate that investments in secure e-government platforms create a "digital infrastructure pathway" that significantly enhances a nation's overall SDG performance. By enforcing mathematical security over policy-based trust, the DTS prototype acts as a localized form of DPI, providing a scalable, tamper-evident foundation for public sector innovation that meets global sustainability and accountability targets.

2.2 Related Studies and Systems

This section evaluates applied research and existing DTSs, distinguishing between foreign architectural innovations and local implementations tailored to the Philippine administrative context.

2.2.1 Foreign Studies and Systems

The global landscape of document tracking has shifted toward high-integrity architectures that integrate cryptographic ledgers within centralized database environments. Foreign studies increasingly address the limitations of traditional policy-based security by proposing Mathematical-Based alternatives that satisfy strict administrative and financial compliance rules. These studies serve as the technical vanguard for modernizing e-governance infrastructure through the use of immutable ledgers and automated verification (Bajoria, 2025; Kim & Kim, 2024).

A synthesis of international research identifies a clear trend toward scalable, process-specific security models. Kim and Kim (2024) demonstrated that adaptive hash chains can provide real-time auditability without the computational bottlenecks typically associated with global blockchains. This is complemented by the work of Bajoria (2025), who proved that blockchain-inspired audit trails can be successfully implemented within standard Relational Database Management Systems (RDBMS). Furthermore, the long-term success of the U.S. Federal Courts' CM/ECF system (Greenwood and Bockweg, 2012) and the digital transformation of the Malaysian (Saman & Haider, 2012) and Indonesian (Zulaeha, 2023) judiciaries underscore the importance of centralized repositories in achieving paperless transactions and immediate document access. However, as noted by Zhai et al. (2022) and Al-khafaji et al. (2020), these repositories require advanced authority authentication to ensure the authenticity and objectivity of historical data.

These foreign studies help this thesis by establishing the architectural benchmarks for the Trust Builder module. The work of Kim and Kim (2024) specifically informs the

micro-sharding design of the DTS prototype, allowing for fast verification even in high-volume environments. By adopting the principles of high-integrity centralized ledgers proposed by Bajoria (2025) and Mishra and Ganesan (2025), the researchers can justify the use of mathematical proofs over simple database logs. These studies provide the technical validation needed to claim that a centralized, LAN-based system can achieve blockchain-level security without the need for decentralized consensus.

2.2.2 Local Studies and Systems

In the Philippines, the development of DTSSs is heavily influenced by the legal mandates of the Ease of Doing Business and Efficient Government Service Delivery Act (Republic Act No. 11032). Local research focuses on the practical application of synchronization technologies, such as QR codes and barcodes, to improve administrative transparency and accountability within government units and higher education institutions (Acedo et al., 2025; Piloton et al., 2023).

The synthesis of local studies reveals a strong emphasis on user acceptability and the digitization of multi-stage workflows. Acedo et al. (2025) and Farin (2024) have proven that QR-based tracking is highly accepted in the Philippine academic and government sectors due to its simplicity and effectiveness in providing verifiable timestamps. Other researchers, such as Acoba (2024), Raralio et al. (2025), and the developers of Castro et al. (2022), have focused on specialized environments like judicial courts and procurement offices, demonstrating the utility of central repositories for case management and purchase requests (Rellon et al., 2020). Furthermore, studies by Trinidad (2022) and Olalia and Bugayong (2014) highlight the importance of digital repositories for institutional accreditation, where the management of thousands of exhibits is a critical requirement (Giraldo, 2016). Despite these successes, local systems often rely on database-level logs that remain vulnerable to silent tampering by administrators.

These local studies help the making of this thesis by identifying the specific operational constraints and user preferences within the Philippine bureaucracy. The high acceptability of QR codes documented by Farin (2024) and Piloton et al. (2023) justifies their use as the primary synchronization mechanism in the DTS prototype. By analyzing the workflow gaps in existing systems like CatSU's DTS (Acedo et al., 2025) and J-CIMS (Acoba, 2024), the researchers can focus on the unique contribution of this project: adding a layer of cryptographic non-repudiation that these local systems currently lack. This ensures

that the prototype is not only compliant with RA 11032 but also addresses the data integrity risks identified in the local literature.

2.2.3 Table of Related Systems

Author/Year	Title of Study/System	Location	Features	Strengths	Limitations	Relevance
Kim & Kim (2024)	Data-Tracking in Blockchain Utilizing Hash Chain: A Study of Structured and Adaptive Process	Foreign	Independent hash chains for specific process lines.	Low computational cost; fast tracking and verification.	Not yet applied to centralized document management.	Supports the Micro-sharding design used in the DTS for scalability.
Bajoria (2025)	Blockchain-Enabled Audit Trails: Revolutionizing RDBMS Data Integrity and Security	Foreign	Mathematically linked database operations in SQL.	Satisfies strict compliance without decentralized networks.	Focuses primarily on financial ledgers.	Validates the concept of building a “High-Integrity Centralized Ledger.”
Mishra & Ganesan (2025)	Securing E-governance: a blockchain-based framework for tamper-proof PDF document exchange	Foreign	Tamper-proof document exchange via blockchain.	High security for shared external documents.	High computational overhead for local hardware.	Highlights the need for a lightweight, centralized alternative.
Acedo et al. (2025)	Web-Based Document Tracking System for Catanduanes State University with QR-Code Technology: In Alignment	Local	QR scanning for real-time status tracking.	Directly aligns with RA 11032 mandates.	Lacks mathematical protection against retroactive tampering.	Guides the implementation of QR scanning at hand-offs.

	with Republic Act No. 11032					
Piloton et al. (2023)	Development of Records Tracking Management System with QR Code	Local	Digital accountability via physical scanning.	Improves visibility into document location.	Relies on policy-based security (database-only logs).	Establishes the standard for physical-digital synchronization.
Acoba (2024)	Judicial Court Information Management System (J-CIMS) for Regional Trial Court in the Philippines	Local	Case filing and status monitoring for courts.	Digitizes complex, multi-stage judicial workflows.	Lacks cryptographic non-repudiation features.	Informs the design of multi-departmental tracking logic.
Farin (2024)	Acceptability and Usability of Quick Response Code for On Line Document Tracking in a Higher Education Institution in the Philippines	Local	Usability assessment of QR-based tracking.	Demonstrates high user acceptance of QR technology.	Focuses on usability rather than mathematical security.	Justifies the use of QR codes for the DTS prototype.

2.3 Synthesis of the Review

The synthesis of the reviewed literature and studies establishes a clear technical progression from manual, paper-based tracking to basic digital repositories. The literature consistently identifies the inherent failures of manual logbooks, such as systemic misplacement (Ozuruonyeoke et al., 2025), information opacity (Maina et al., 2020), and legal non-compliance with turnaround mandates (RA 11032, 2018). Foreign studies on high-integrity judicial systems (Saman & Haider, 2012; Zulaeha, 2023) and the U.S. Federal Courts (Greenwood and Bockweg, 2012) demonstrate that while centralized digitization significantly improves case throughput, it remains vulnerable to data corruption and unauthorized modifications without advanced authentication layers (Al-khafaji et al., 2020; Zhai et al., 2022).

The transition toward secure architectures is characterized by the adoption of mathematical-based security measures that ensure non-repudiation and tamper-evidence. Scholarly works on hash-chaining (Crosby & Wallach, 2009; Kim & Kim, 2024) and digital signatures (Bernstein et al., 2012; Chavarro et al., 2018) prove that security must be embedded directly into the data structure rather than relying on supervisor oversight. Furthermore, the integration of synchronization technologies like QR codes (Acedo et al., 2025; Farin, 2024) and NFC/IoT (Abidin et al., 2020) provides the necessary bridge between physical file movement and digital records. This ensures that the digital trail accurately reflects the ground truth of administrative actions, fulfilling the accountability requirements of modern governance (Acedo et al., 2025).

From a socio-technical perspective, the success of these systems depends on competing with the perceived convenience of Shadow Systems and unofficial workflows. Research into user behavior identifies shadow systems, such as Google Sheets, as rational coping mechanisms for rigid IT environments (Behrens, 2009; Šedivcová & Potančok, 2019). However, the literature warns that these unapproved tools introduce severe security risks and fragmented databases (Haag & Eckhardt, 2024). Consequently, for a DTS to be effective, it must integrate human-centric design principles (Aqeel et al., 2023) and decision-support algorithms (Keen & Scott Morton, 1978; Ngan et al., 2022) to provide a user experience that is both more secure and faster than the informal alternatives (Lingaya, 2019).

Despite these advancements, a critical research gap remains in the implementation of mathematical-based security within local, resource-constrained government environments. Most existing local systems in the Philippines rely on Policy-Based Security, where the integrity of document logs depends on the hope that administrators will follow rules rather than on mathematically unbreakable proofs (Bajoria, 2025; Chavarro et al., 2018). While international studies propose decentralized blockchain solutions to solve this, these architectures are often too computationally expensive and complex for the local area networks (LAN) typical of provincial government offices (Mishra & Ganesan, 2025). While Mishra and Ganesan (2025) advocate for decentralized exchange, this study prioritizes the centralized hash-chain model proposed by Bajoria (2025). For a single-office environment like DepEd Iligan, a centralized high-integrity ledger provides the same mathematical immutability as a blockchain without the energy overhead or latency of network-wide consensus, making it more sustainable for local government hardware. Furthermore, current local systems often lack a unified intake protocol that enforces a single primary key for every document from intake to release (Acedo et al., 2025).

This study addresses these gaps by implementing a high-security Trust Builder module that provides blockchain-level integrity within a centralized, LAN-based environment. By integrating SHA-256 hash-chaining (Crosby & Wallach, 2009) and Ed25519 digital signatures (Bernstein et al., 2012), the proposed system shifts the source of trust from human supervisors to verifiable mathematical proofs. The system's Active Guard and *Auto-Freeze* module provide a real-time defense against the silent tampering identified in the literature (Zhai et al., 2022), while its smart routing and QR-sync features ensure it matches the speed of informal shadow systems (Behrens, 2009). This provides DepEd Iligan City with a specialized security artifact that enforces institutional accountability and prevents the proliferation of unofficial tracking methods.

2.4 Theoretical Framework

The development of the DTS prototype is grounded in a multi-disciplinary theoretical framework that bridges the gap between administrative case management and cryptographic engineering. These theories serve as the *Knowledge Base* for the Design Science Research (DSR) methodology, providing the necessary rigor to justify the architectural decisions behind the system's core modules (Hevner, 2007). By anchoring the artifact in established theories, the research ensures that the Trust Builder and Active Guard modules are not merely technical features but are evidence-based solutions to identified institutional problems.

The primary foundation of the system is Information Integrity Theory, which focuses on maintaining the accuracy, consistency, and reliability of data over its entire lifecycle. This theory provides the justification for implementing Mathematical-Based Security through SHA-256 hash-chaining and Ed25519 digital signatures (Bajoria, 2025; Bernstein et al., 2012). It shifts the source of institutional trust from policy-based human oversight to verifiable mathematical proofs, ensuring that document logs are tamper-evident and non-repudiable. Furthermore, Case Management Theory (Frankel et al., 2018) provides the structural logic for monitoring the document through its *Intake*, *Processing*, and *Releasing* stages. This theoretical lens allows the system to accurately digitize the physical movement of correspondence while maintaining a strict audit trail of departmental responsibility.

To address the operational challenges of high-volume routing and physical synchronization, the study incorporates Decision Support Theory (Keen & Scott Morton, 1978) and General System Theory (Bertalanffy, 1968). Decision Support Theory informs the development of the heuristic route prediction module, which utilizes learned keyword weights to help Records Officers manage routing tasks efficiently (Ngan et al., 2022).

Meanwhile, General System Theory views the DepEd Division Office as a socio-technical system where technological tools must harmonize with human habits. This theory justifies the integration of QR code scanning during physical hand-offs, ensuring that the digital status updates are perfectly synchronized with the physical location of the files (Bertalanffy, 1968; Acedo et al., 2025). Together, these theories provide the comprehensive framework needed to design, implement, and evaluate the technical integrity of the DTS artifact.

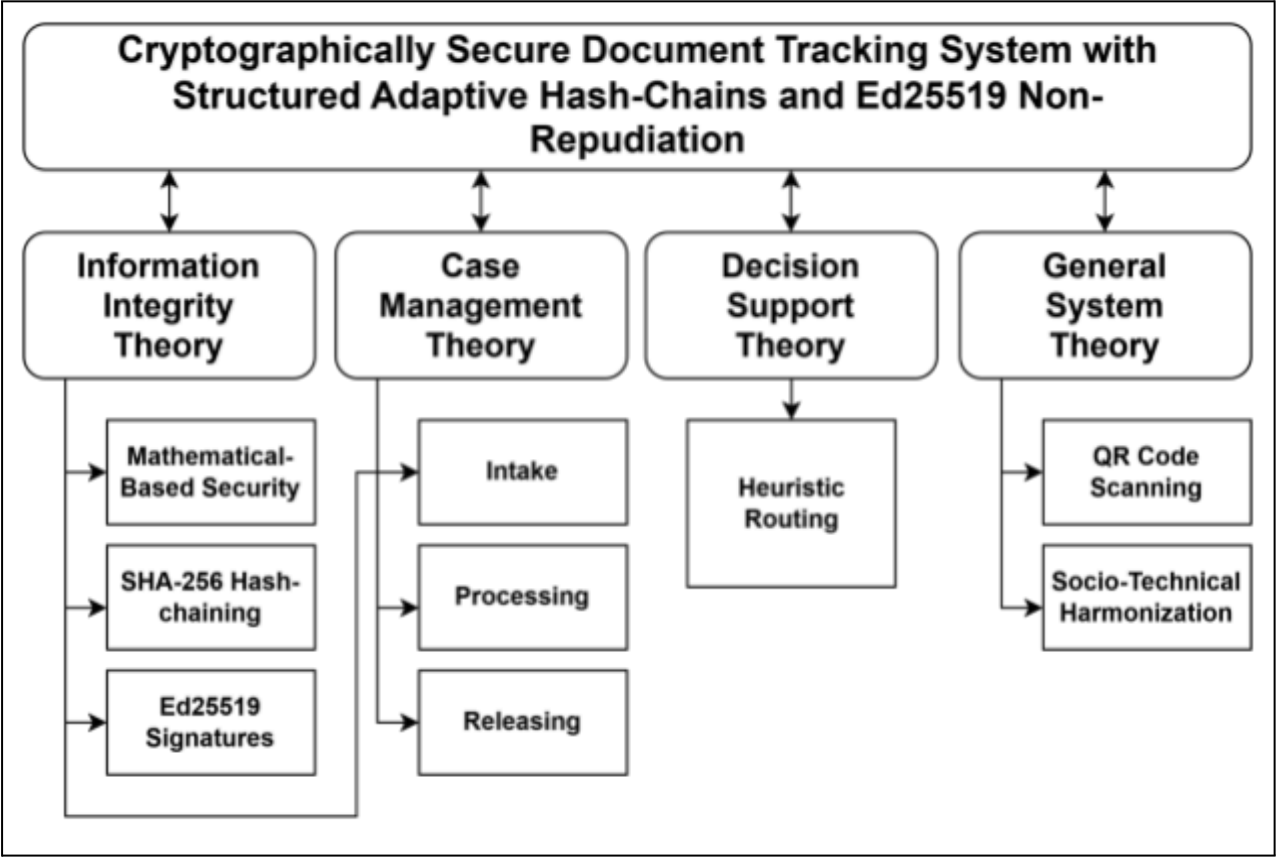


Figure 2.4: Theoretical Framework Diagram of Cryptographic DTS

2.5 Conceptual Framework

The conceptual framework of this study is operationalized through an Input-Process-Output (IPO) model, which illustrates the logical flow and transformation of data within the DTS prototype. This model serves as the blueprint for translating the theoretical foundations of the study into a functional IT artifact. By mapping the system’s technical requirements to its organizational outcomes, the conceptual framework provides a structured approach to solving the identified gaps in document accountability and data integrity.

Input: The system intake begins with the collection of document metadata (Title, Purpose, and Submitter Information), user authentication credentials, and security initialization data. For cryptographic operations, the system requires user-provided Security PINs to derive encryption keys. Additionally, the system inputs include departmental route definitions and real-time scanning data generated via QR codes during physical document transfers.

Process: The central processing phase is managed by the Trust Builder module, which performs SHA-256 hash-chaining and Ed25519 signature generation to ensure mathematical non-repudiation. Simultaneously, the Active Guard middleware serves as a real-time integrity monitor, verifying document states before authorizing any movement. The system also incorporates Decision Support logic through TF-IDF-inspired route prediction and employs asynchronous background processing to handle high-volume analytics and system-wide integrity audits.

Output: The primary outputs of the DTS include a tamper-evident, immutable audit trail for every document and cryptographically signed logs that provide verifiable proof of administrative actions. The system also delivers real-time status visibility for both guests and personnel, automated integrity reports for system administrators, and bottleneck analytics. Collectively, these outputs ensure full compliance with the transparency and efficiency mandates of Republic Act No. 11032.

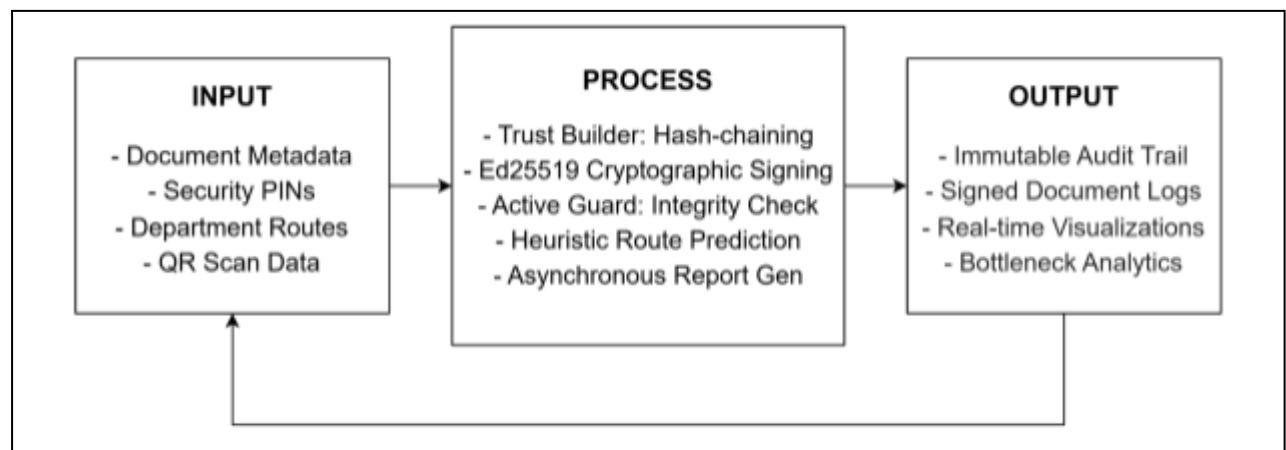


Figure 2.5: Conceptual Framework for Cryptographic DTS using IPO Model

CHAPTER 3

RESEARCH METHODOLOGY

This chapter details the research design and development methodology used to create the DTS for DepEd Iligan City. The study follows the Design Science Research (DSR) methodology to address institutional problems through the creation of an innovative IT artifact, utilizing Rapid Application Development (RAD) as the strategy for building and refining the system.

3.1 Research Design (DSRM)

The researchers employ the Design Science Research (DSR) methodology. This focuses on both the creation of purposeful artifacts and the generation of design knowledge (Hevner et al., 2004). The study adopts the Design Science Research Methodology (DSRM) proposed by Peffers et al. (2007). This provides a systematic process for developing and evaluating artifacts in Information Systems research.

3.1.1 DSR Cycles (Hevner, 2007)

DSR Cycle	Activity
1. Relevance Cycle	Identify problems: Race conditions in kiosks and “Shadow System” usage. Collect requirements from Records and IT Department Staff.
2. Design Cycle	Build the DTS prototype with the Trust Builder, Active Guard modules and QR hand-off logic. Refine the artifact iteratively based on Records and IT Department Staff feedback.
3. Rigor Cycle	Use existing knowledge: SHA-256 hash chains (Mishra and Sato, 2025) and DSRM stages to support design decisions.

3.2 Research Framework

The study is guided by the DSR framework adapted from Hevner et al. (2004), which connects the organizational environment, design science activities, and the scientific knowledge base. This framework ensures that the DTS prototype is not merely a technical

solution but a rigorous response to the administrative and security gaps identified at the DepEd Division Office of Iligan City.

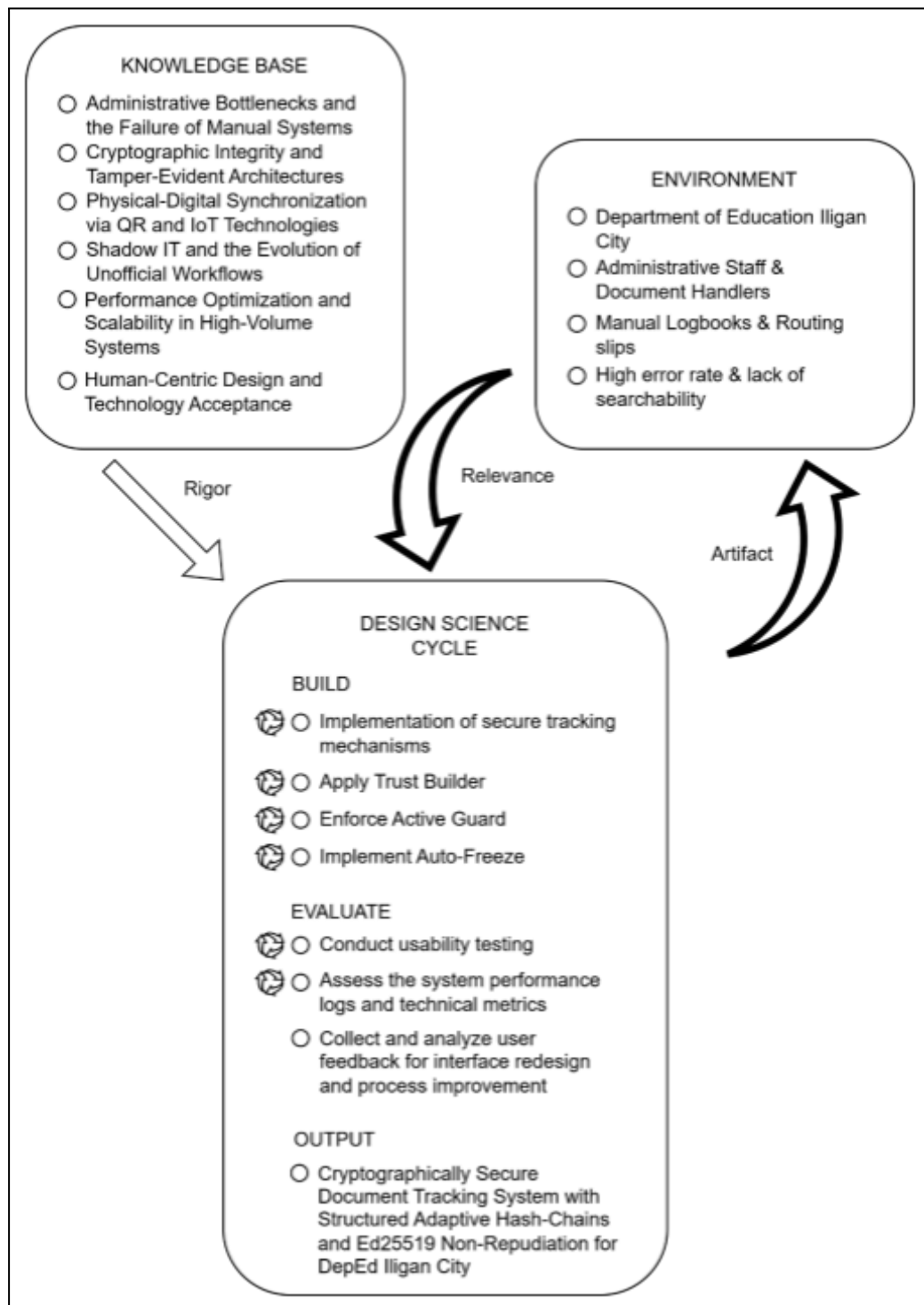


Figure 3.1: Design Science Research Framework adapted from Hevner et al. (2004)

By utilizing this framework, the researchers maintain a continuous *Build-Evaluate* loop. This iterative process allows for the refinement of the system's performance

layer—such as the TF-IDF-inspired route prediction (Ngan et al., 2022) and asynchronous background processing—until it effectively meets the institutional demands for speed and absolute data integrity. This approach guarantees that the final artifact is methodologically sound and operationally viable within the local-area network (LAN) constraints of the division office.

3.3 Problem Identification and Motivation

The motivation for this study stems from the critical technical failures and operational inefficiencies observed at the DepEd Division Office of Iligan City. During an empirical visit in December 2025, researchers identified a significant race condition at guest kiosks, caused by a lack of robust queue management (Silberschatz et al., 2020). This failure resulted in race conditions where concurrent document submissions would overwrite one another, leading to lost tracking data and administrative backlog. This technical instability directly undermines the transparency mandates of the Ease of Doing Business Act (Republic Act No. 11032).

More significantly, due to empirical observation the study is motivated by the pervasive use of informal Shadow Systems within the institution. Because the formal tracking system was perceived as incomplete or rigid, the Records Department resorted to using unencrypted Google Sheets and Forms to log metadata for over 10,000 document entries. As noted by Behrens (2009) and Rentrop and Zimmermann (2020), such shadow systems emerge as rational coping mechanisms for staff but introduce severe cybersecurity risks and lack the ACID compliance needed for government data sovereignty (Haag & Eckhardt, 2024). The creation of an innovative, high-security IT artifact is therefore necessary to consolidate these fragmented audit trails and restore institutional trust through mathematical-based assurance.

3.4 Objectives of the Solution Table

Based on the identified institutional gaps, the study establishes clear objectives for the proposed DTS prototype. These objectives serve as the measurable criteria for evaluating the success of the artifact across functional, non-functional, technical, and theoretical dimensions:

Objective Type	Description
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Functional Objectives	Enforce a mandatory intake protocol to systematically mitigate ghost documents, provide heuristic route suggestions (Ngan et al., 2022) as a secondary heuristic tool for efficiency, and implement QR-sync for hand-offs (Acedo et al., 2025).
Non-Functional Objectives	Ensure sub-second response times for dashboard navigation and document searches, maintaining system responsiveness even when managing over 1,000,000 records (Silberschatz et al., 2020).
Technical Objectives	Implement the Trust Builder module using SHA-256 hash-chaining (Crosby & Wallach, 2009) and Ed25519 for digital signatures (Bernstein et al., 2012) due to its superior performance and smaller 256-bit key size compared to standard RSA. The TF-IDF route prediction module is implemented as a heuristic decision-support tool rather than an autonomous classifier; it presents weighted keyword-based routing suggestions to the Records Officer, who retains full authority over final routing decisions. This approach is consistent with the decision-support framework proposed by Ngan et al. (2022), where algorithm-generated recommendations augment rather than replace human judgment. Use Logical Micro-sharding (table partitioning) to isolate chains and ensure sub-second verification. Integrate an Active Guard module for real-time state verification.
Theoretical Objectives	Demonstrate how Mathematical-Based Security and high-integrity centralized ledgers can be practically implemented in a resource-constrained government environment (Bajoria, 2025; Kim & Kim, 2024).

3.5 Development Methodology: Rapid Application Development (RAD)

While DSRM provides the research wrapper, the researchers engineered the artifact using RAD (Martin, 1991) across four iterative phases. This approach translated abstract cryptographic concepts into a tangible system. It allowed for continuous feedback from the Records and IT Department during the construction of the prototype.

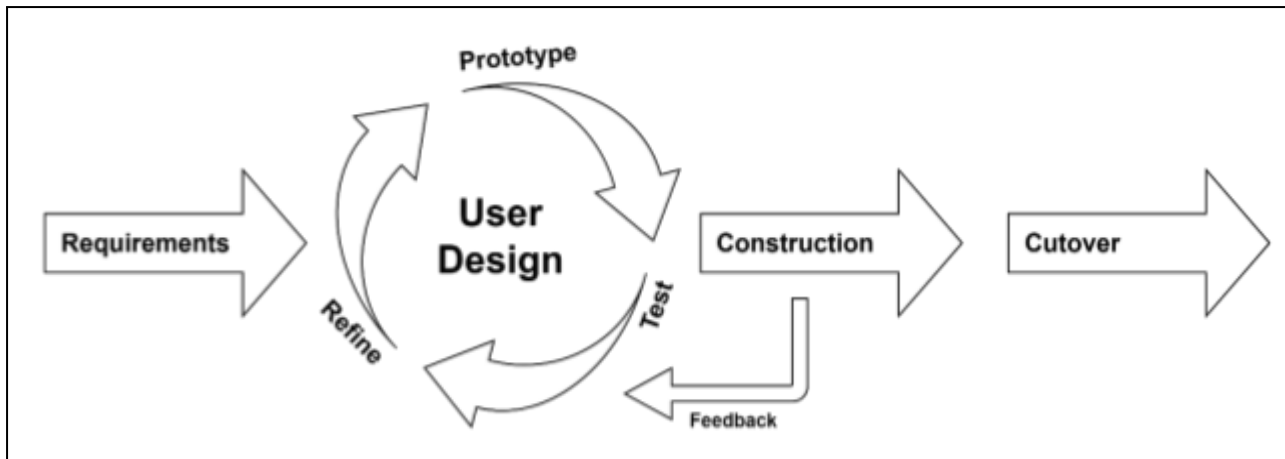


Figure 3.5: Design Science Research Framework adapted from Hevner et al. (2004) with Rapid Application Development

3.6 Research Locale and Participants

The study is conducted within the DepEd Division Office of Iligan City, specifically focusing on the administrative workflow of official documents across various functional units (Records, Cash, Personnel, Supply, etc.). This environment was selected because it represents a high-volume government ecosystem where data integrity and physical accountability are critical for operational transparency.

The initial needs assessment was conducted through verbal consultations with key personnel to identify the operational gaps in the existing system. During these discussions, the researchers shared their first-hand experiences regarding the race condition failures at the guest kiosks. The Records Department confirmed these technical instabilities and further disclosed systemic faults in the current DTS that forced the staff to resort to unofficial shadow systems, such as Google Forms and Google Sheets, to ensure operational continuity.

Informed consent was obtained from all participants of this verbal assessment, with the researchers proposing to use these identified institutional challenges as the foundation for their thesis project—a proposal that was encouraged by the Records Department personnel.

3.7 Evaluation Framework

While the DSRM (Peppers et al., 2007) governs the overall research process — from problem identification through artifact creation — the evaluation phase is specifically structured using the Framework for Evaluation in Design Science (FEDS) proposed by Venable et al. (2016). FEDS operates within the DSRM's Rigor Cycle as a dedicated

instrument for assessing artifact utility, quality, and efficacy, providing a more granular evaluation structure than DSRM prescribes alone. The evaluation strategy incorporates both ex-ante (predictive) and ex-post (empirical) components to assess the prototype’s resilience and operational readiness:

Evaluation Type	Setting	Method	Purpose
Ex-ante	Artificial	Automated Integrity Simulation: Recalculating 10,000+ document hashes and verifying signatures against historical keys using the IntegrityCheckJob. Performance Load Testing: Measuring sub-second response times under concurrent database load.	Assess expected performance, cryptographic resilience, and algorithmic scalability before full institutional deployment.
Ex-post	Naturalistic	One-Week Field Implementation: Deploying the DTS as the primary system at kiosks and offices. System Usability Scale (SUS): Standardized usability surveys conducted with both institutional staff and public clients (guests).	Assess real-world effectiveness, interface clarity, and institutional acceptance under actual operational conditions.

This dual-layered approach ensures that the Trust Builder module is not only mathematically sound but also practically usable during high-traffic scenarios.

3.8 Data Gathering Procedures

Data is gathered from a combination of live technical logs and human-centric feedback instruments during a one-week implementation phase to provide a comprehensive measure of the system’s success in a real-world environment:

One-Week Live Implementation: With the support of the Records Department, the DTS prototype is deployed as the primary tracking system at the guest intake kiosks and within the internal departments for exactly one week. This naturalistic deployment captures real-world usage data, including concurrent guest submissions and department-to-department QR hand-offs.

System Performance Logs and Technical Metrics: During the live week, quantitative data is extracted directly from the system’s database metrics, including average query times,

connection counts, and the occurrence of race conditions. The IntegrityCheckJob runs in the background to provide metrics on the *Verified Percentage* of the live hash-chain.

Data Export and Schema Restoration: At the conclusion of the one-week deployment, all gathered document logs and metadata are exported. By acquiring the database schema of the existing institutional DTS, the exported data is formatted to seamlessly integrate back into their current system, allowing the original DTS to safely take over operations without data loss.

System Usability Scale (SUS) Questionnaire: Usability testing is conducted with two distinct user groups during the live implementation: public clients (guests) using the intake kiosks, and internal staff/Records Officers managing the routing. Both groups complete a digital questionnaire based on the SUS (Brooke, 1996) immediately after interacting with the system.

Key Informant Interviews: Informal post-implementation interviews are conducted with IT personnel to gather qualitative insights into the scalability of the micro-sharding hash-chain design and the robustness of the Active Guard module.

All data gathering instruments and technical protocols are designed to adhere to the rigor cycle of the DSR methodology, ensuring that the findings are supported by empirical evidence and verifiable system logs.

3.9 Data Analysis

Data collected during the one-week live implementation is processed and analyzed using both quantitative and qualitative methods to ensure a holistic evaluation of the DTS artifact:

Quantitative SUS Score Analysis: The results of the System Usability Scale (SUS) surveys from both clients and staff are calculated to determine standardized usability scores (0 to 100). The research sets a target rating of Good (a score greater than 68) to validate that both the public-facing kiosk interface and the internal cryptographic routing interfaces are accessible.

Operational and Performance Metric Verification: Technical logs from the live deployment are analyzed to verify the complete mitigation of kiosk race conditions under real-world concurrent load — specifically, simultaneous guest submissions recorded at the intake kiosks during operational hours. System performance is benchmarked against the

actual hardware utilized during the field implementation — specifically a local Linux-based server environment (the researcher's laptop, equipped with a 6-core processor and 16GB of RAM). Laravel's built-in query logging and the system's database performance metrics serve as the primary instruments for capturing average query execution times, connection counts, and response durations, with a target of maintaining sub-second response times (less than 1,000 milliseconds) during peak usage.

Thematic Analysis of Qualitative Feedback: Feedback from post-implementation interviews with IT personnel undergoes thematic analysis to identify any operational friction points and evaluate the real-world robustness of the Active Guard module.

Integrity Audit Analysis: The results of the live IntegrityCheckJob are analyzed to quantify the system's Verified Percentage in a naturalistic setting. A 100% verified result is required to validate that the SHA-256 hash-chaining and Ed25519 signatures are technically sound.

3.10 Ethical Considerations

The study strictly adheres to the principles of the Republic Act No. 10173 (Data Privacy Act of 2012). Ethical approval was secured from the relevant institutional committee prior to formal data collection, and all research procedures prioritize the protection of institutional and personal information:

Informed Consent: Participation in usability testing and interviews was strictly voluntary. Researchers provided clear orientations regarding the study's objectives, and signed informed consent was obtained from all 10 institutional respondents prior to any evaluation tasks.

Anonymity and Confidentiality: To protect the privacy of the personnel, participants are assigned identification numbers (e.g., Participant 01) in all reports and analysis logs. No personally identifiable information (PII) is stored within the system's technical performance tables.

Data Sovereignty and Network Security: Adhering to the IT Department's strict security mandates, the DTS prototype is deployed exclusively within a local area network (LAN). This on-premise deployment ensures institutional data sovereignty, preventing unauthorized external access to sensitive document metadata and cryptographic keys.

System Integrity: All collected evaluation data is stored on secure, local devices as mandated by the institutional IT policy and is used solely for the evaluation and academic documentation of the DTS prototype.

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